

How UniHelp generalizes — and how an admin sets up a new campus

UniHelp is one runtime, many campuses. IIT Patna is the default profile, not the product. A second residential campus is configured in chat by an authenticated admin: catalog, people, and optional extra services. The seven domain agents stay the same.

1 runtime
Planner → executor → agents

3 layers
Profile · people · domain data

Admin chat
Configure, import, add features

No raw SQL
Model never gets the database

The rule

Same shape of campus (hostels, mess, buses, rooms, timetable) = config + data, not a code rewrite. A hospital or a day college with no hostels is a different product: turn unused agents off, and add a real new agent in code — not a prompt.

1. What stays the same vs what changes

Layer	What it is	Who changes it	IIT Patna example
Runtime	Auth, planner, executor, seven agents, <code>chat(input, metadata)</code>	Engineers. Frozen for a campus rollout.	Same process on port 8002
Org profile	YAML: name, timezone, roll regex, enabled agents, catalog names + aliases	Admin, via setup prompts (or <code>orgs/<id>.yaml</code>)	Kalam, Bus 02, SAC Hall
People	Canonical <code>campus_agent.people</code> (role, external id, auth key)	Admin import prompt, or CSV	<code>student-demo</code> → Aarav Sharma / 2501CS09
Domain rows	Menus, bus trips, bookings, attendance, notices, timetable slots	Admin/faculty through the existing domain agents	“Change Kalam dinner to dal”
Custom features	JSON-backed extra service (library, lost-and-found). Shared tables, not new DDL.	Admin “add a feature” prompt	Optional — none on the default profile

2. How generalization actually works

Requests still authenticate, plan, and dispatch the same way. What changed is *where names come from*.

CATALOG-DRIVEN PARSE

fast_parse

Hostel, bus, room, and meal matches read `ORG.catalog`. Aliases (“arya bhatta”, “CV Raman”) are optional. No match → `None`, never a guess.

AGENT REGISTRY

planner + executor

The seven agents are listed once. The planner prompt and dispatch map are generated from that list, filtered by `enabled_agents`.

ONE IDENTITY TABLE

people

`authenticate()` reads `campus_agent.people`. Domain agents keep receiving the same `user_metadata`. Student ids are checked against the org regex.

The setup agent is **admin-only**. It is always dispatchable, but it is not listed in the student planner prompt, so a student saying “add a feature” is not routed there. Every setup tool re-checks `role == admin`.

What “add a feature” means

The model does not write Python or `CREATE TABLE`. Admin supplies a name, keywords, and a field schema (`string | number | boolean | date`). Rows live in `custom_feature_records` as JSON. The planner then routes e.g. “search the library” to that feature.

3. New admin: first-hour setup (all in chat)

Sign in as admin (`admin-demo` on the seed database, or the key issued at import). The server must already be running with the setup agent loaded (`python main.py` on `127.0.0.1:8002`).

STEP 1

Name the campus

Display name, timezone, roll-number pattern, which of the seven agents are on.

STEP 2

Fill the catalog

Hostels, buses, rooms, meals, complaint categories, plus everyday aliases.

STEP 3

Import people

CSV or a JSON list. Invalid student rolls abort the batch. Generated keys are shown once.

STEP 4

Seed domain data

Use the existing agents: menus, bus trips, notices, timetable slots.

STEP 5

Optional feature

“Add a library...” — JSON service, then smoke-test as a student.

Prompts the admin can paste

CONFIGURE CAMPUS · SETUP AGENT

Rename us to IIT Mandi, timezone Asia/Kolkata. Hostels: Chandra (alias “C hostel”), Suvalsar. Rooms: SAC Hall, Guest House. Buses 01, 02, 03, 04. Meals: breakfast, lunch, snacks, dinner. Complaint categories: academic, hostel, mess. Student rolls match `^[0-9]{4}[A-Z]{2}[0-9]{2}$`. Keep all seven agents enabled.

IMPORT PEOPLE · SETUP AGENT

Import these people as CSV:
`role,name,roll_number`
`admin,Campus Admin,AD001`
`faculty,Priya Patel,PF001`
`student,Asha Rao,2501CS99`

Columns accepted: `role`, `display_name` or `name`, `external_id` or `roll_number`, optional `authentication_key`. At most 500 rows per prompt. Missing keys are generated — copy them out of the reply; they are not shown again.

SEED OPERATIONAL DATA · EXISTING DOMAIN AGENTS

Set today’s Chandra breakfast to poha and tea.
Add Bus 01 weekdays 08:00 from Chandra to the academic block, driver Test, phone 9999999999.
Publish a notice: “Orientation Monday 9am, Auditorium.”

ADD A FEATURE · SETUP AGENT

Add a library feature. Keywords: library, book, isbn. Fields: title (string, required), author (string, required), isbn (string, optional). Students can search books and add a hold.

SMOKE-TEST AS A STUDENT

What’s today’s breakfast at Chandra?
What is the bus schedule for Bus 01?
Search the library for algorithms.

4. What the admin prompt will not do

Request	What actually happens
“Run this SQL” / “create a fees table”	Rejected. Setup has no SQL tool. Invalid org YAML is refused before the file is replaced (previous copy kept as <code>.bak</code>).
“Add a hospital agent with its own schema”	Not a JSON feature. That needs a real agent in the registry — engineering work.
Empty catalog (no hostels, no buses)	Startup and save fail loud. A missing field must not silently disable a route.
Student or faculty trying setup	Planner does not offer setup. The agent itself also refuses.
Reuse a built-in name (<code>mess</code> , <code>bus</code> , ...)	<code>add_feature</code> returns an error. Reserved slugs cannot be overwritten.

5. Built-in agents the admin can leave on or turn off

Agent	Student can	Admin/faculty can also
mess	Read menus	Change menus
bus	Read schedules	Add / remove / edit trips
complaint	File and list	Verify and complete
room_booking	Check / request	Approve, reject, book directly
attendance	Own percentages	Mark, courses, enrollments
notice	Read board	Publish; admin can archive
timetable	Own classes	Add / update / delete slots

Disable with: "Turn off mess and bus; we are a day college." `enabled_agents` must still name agents that exist in this repo.

Checklist before you hand it to students

- 1. Admin prompt updated display name and catalog; a hostel alias resolves.
- 2. People imported; at least one student, faculty, and admin key works.
- 3. One live query per enabled agent returns campus-correct names, not IIT Patna leftovers.
- 4. If a custom feature was added, a student query using its keyword hits that feature, not setup.
- 5. Generated authentication keys stored offline; they are not re-printed.

If YAML was edited on disk

Admin can say "Reload the org profile." `update_org_profile` and `add_feature` already reload routing. Reload is only for a manual file edit.

Do not skip validation

A bad regex or empty hostel list fails the save. That is intentional: silent misconfig is how a campus "loses" a hostel.